Curve tracing

* Procedure of tracing cartesian curve

The following steps are useful to trace the curve equation $f(x,y) = 0$

→ Step 1 : Symmetry

A curve is symmetric about

(a) x-axis if equation contains only even powers of y

Ex: Parabola $y^2 = 4ax$

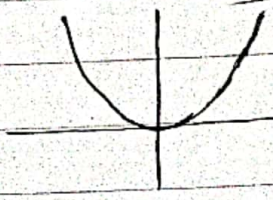


Ex: $x^2 + y^2 = ax$



(b) y-axis if equation contains only even powers of x.

Ex: $x^2 = 4ay$

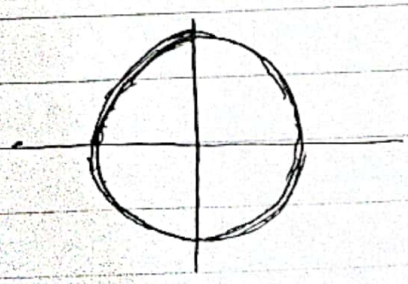


Ex: $x^2 + y^2 = ay$



(c) both the axes if equation contains even powers of both x & y.

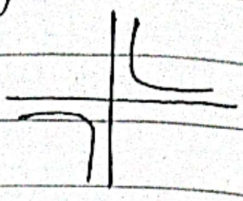
Ex: $x^2 + y^2 = a^2$



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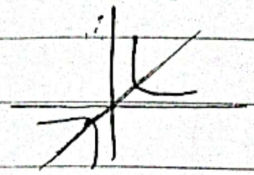
(d) in opposite quadrant if on replacing x & y by $-x$ & $-y$ respectively, equation remains unchanged.

Ex: Hyperbola $xy = a^2$

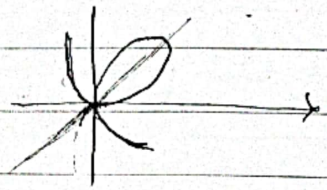


(e) the line $y = x$ if on interchanging x & the equain remains unchanged.

Ex: Hyperbola $xy = a^2$



Ex: $x^3 + y^3 = 3axy$



→ Step 2 : Origin

(a) The curve passes through the origin if x & $y = 0$ satisfy the equation.

Ex: Parabola $y^2 = 4ax \Rightarrow y^2 - 4ax = 0$

If we put $x = 0$ & $y = 0$, then eq^n is satisfied. ∴ Curve passes through the o

(b) If curve passes through the origin, find equations of tangent at origin are obta by lowest degree terms = 0

For Ex: The curve $y^2(2a - x) = x^3$ passes through the origin. Curve eq^n $2ay^2 - xy^2 - x^3 = 0$

To find tangent at the origin.

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Formula : lowest degree terms = 0

$$2ay^2 = 0$$

$$\Rightarrow y^2 = 0 \Rightarrow y = 0, 0 \text{ (x-axis)}$$

are tangent at the o

Ex: Find the tangent at the origin to the curve $y^2(a+x) = x^2(3a-x)$

Soln:

$$y^2(a+x) = x^2(3a-x)$$

$$\Rightarrow ay^2 + xy^2 - 3ax^2 + x^3 = 0$$

Here $x = 0, y = 0$ satisfy the given equation

∴ Given curve passes through the origin.

→ Tangent at the origin

Formula : lowest degree terms = 0

$$ay^2 - 3ax^2 = 0$$

$$\Rightarrow ay^2 = 3ax^2$$

$$\Rightarrow y = \pm\sqrt{3}x$$ are tangents at the origin.

(c) Write the nature of origin

(i) Tangents are real & distinct — node

(ii) " " real & coincide — cusp

(iii) " " imaginary — conjugate point

Ex: curve $y^2(a^2+x^2) = x^2(a^2-x^2)$ passes through the origin.

$$a^2y^2 + x^2y^2 - a^2x^2 + x^4 = 0$$

To find the tangent at the origin

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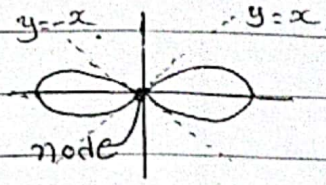
→ lowest degree terms = 0

$a^2y^2 - a^2x^2 = 0$

$\Rightarrow y^2 = x^2$

$\Rightarrow y = \pm x$ — tangents are real & distinct

∴ origin is node



Ex: The curve $y^2(2a-x) = x^3$ passes through the origin

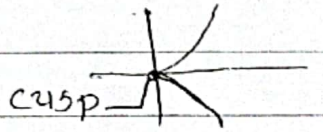
∴ Tangent at the origin

→ lowest degree terms = 0

$2ay^2 = 0 \Rightarrow y^2 = 0$

$\Rightarrow y = 0, 0$ — tangents are real & concide.
(x-axis)

∴ origin is cusp.



Ex: curve $y^2(x-a) = x^2(x+a)$

→ Tangent

lowest degree terms = 0

$-ay^2 = ax^2$

$\Rightarrow y^2 = -x^2$

$\Rightarrow y = \pm ix$ — tangents are imaginary

∴ origin is conjugate point.

→ Step 3 : Point of intersection

(a) For pts. on x-axis, put $y = 0$ ∴ pt. $(x, 0)$

(b) For pts. on y-axis, put $x = 0$ ∴ pt. $(0, y)$

(c) For pts. on $y = x$ line, put $y = x$ in the eqⁿ if curve is symmetric about line $y = x$.

(d) Tangent at the point of intersection is obtained by shifting the origin to this point and then equating the lowest degree term to zero.

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For Ex: The circle $x^2 + y^2 = a^2$

→ For pts. on x-axis, put $y = 0$

$\therefore x^2 = a^2 \Rightarrow x = \pm a$

∴ curve intersects the x-axis at pts. $(a, 0), (-a, 0)$.

→ For pts. on y-axis, put $x = 0$

$\therefore y^2 = a^2 \Rightarrow y = \pm a$

∴ curve intersects the y-axis at pts. $(0, a), (0, -a)$

Ex: The curve $xy = a^2$ is symmetric about the line $y = x$.

∴ For pts. on $y = x$ line, put $y = x$ in given eqⁿ. we get, $x^2 = a^2$

$\Rightarrow x = \pm a$

we get pts. $(-a, -a)$, (a, a)

→ Step 4: Asymptotes

(a) Asymptote parallel to x-axis

Formula: Coefficient of highest power terms of $x = 0$

(b) Asymptote parallel to y-axis

Formula: Coefficient of highest power terms of $y = 0$

(c) Oblique Asymptote: $y = mx + c$

(i) Put $x = 1$ & $y = m$ in the highest degree terms of $f(x, y) = 0$, denoted by $\phi_n(m)$

(ii) Solve $\phi_n(m) = 0$ to determine m.
(Take only real roots.)

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(iii) Find 'C' by formula

$$C = -\frac{\phi_{n-1}(m)}{\phi_n'(m)}$$

Hence, the oblique Asymptote : $y = mx + C$

Ex: Find the asymptotes of the curve $x^2y^2 - x^2y - 5xy^2 + x + y + 5 = 0$

Sol^n:

For asymptotes parallel to x-axis

→ Coefficient of highest power terms of $x = 0$

$y^2 - y = 0$

$\Rightarrow y(y-1) = 0$

$\Rightarrow y = 0, y = 1$ are asymptotes parallel to x-a

For asymptotes parallel to y-axis

→ Coefficient of highest power terms of $y = 0$

$x^2 - 5x = 0$

$\Rightarrow x(x-5) = 0$

$\Rightarrow x = 0, x = 5$ are asymptotes parallel to y-

Ex: Find the asymptotes of the curve $y^3 - x^2(6-x) =$

Sol^n:

Curve $y^3 - x^2(6-x) = 0$

$\Rightarrow y^3 - 6x^2 + x^3 = 0$

Here curve has no asymptotes parallel to the axes.

Let oblieque asymptote : $y = mx + C$

Highest degree = $n = 3$

Highest degree terms = $y^3 + x^3 = 0$.

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put $x = 1$, $y = m$

$\therefore \phi_3(m) = y^3 + x^3 = m^3 + 1$

$\phi_3'(m) = 3m^2$

$\phi_2(m) = -6x^2 = -6 \quad (\because x = 1)$

Now, to find m

Take $\phi_3(m) = 0$

$\Rightarrow m^3 + 1 = 0$

$\Rightarrow (m+1)(m^2 - m + 1) = 0$

$\Rightarrow m = -1$ & other roots are imaginary

$$C = -\frac{\phi_2(m)}{\phi_3'(m)} = \frac{-(-6)}{3m^2} = \frac{6}{3(-1)^2} = 2$$

Hence oblique asymptote : $y = mx + C$

$\Rightarrow y = -x + 2$

→ Step 5 : Region where no part of the curve lies

(a) If it is possible to express the equation as $y = f(x)$ and if y becomes imaginary for some value of $x > a$ (say), then no part of the curve exists beyond $x = a$.

(b) Ily, if it is possible to express the equation as $x = f(y)$ and if x becomes imaginary for some value of $y > b$ (say), then no part of the curve exists beyond $y = b$.

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For Ex: $a^2x^2 = y^3(2a-y)$

$$x^2 = \frac{y^3(2a-y)}{a^2}$$

(i) For $y > 2a$, x becomes imaginary

∴ The curve doesn't exist beyond $y = 2a$

(ii) For negative values of y, x becomes imagi

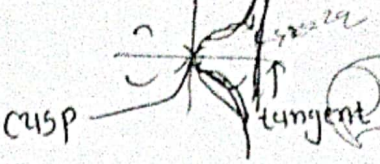
∴ The curve doesn't exist in 3rd & 4th quadrant.

Ex: $y^2 = 4x$

For negative value of x, y becomes imagina

∴ The curve doesn't exist in second & Third quadrant. //

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Examples :

[1] Trace the curve $y^2(2a-x) = x^3$

Sol^n :

We have, $y^2 = \frac{x^3}{(2a-x)}$ —— (i)

$\Rightarrow 2ay^2 - xy^2 - x^3 = 0$ —— (ii)

(1) Symmetry :

The eq^n contains only even powers of y.

∴ Curve is symmetric about x-axis.

(2) Origin : eq^n (ii) is satisfied by $x=0, y=0$

[If we put $x=0, y=0$ in eq^n (ii), then eq^n (ii) is satisfied.]

∴ Curve passes through the origin

→ Tangent ~~at~~ the origin

lowest degree terms $= 0$

$\Rightarrow 2ay^2 = 0 \Rightarrow y^2 = 0$

$\Rightarrow y = 0, 0$ — Tangents are real & coincide.
(x-axis)

∴ origin is cusp.

(3) Points of intersection

→ For pts. on x-axis, put $y=0$ in eq^n (ii)

∴ $-x^3 = 0 \Rightarrow x = 0$ ∴ pt $(0,0)$

→ For pts. on y-axis, put $x=0$ in eq^n (ii)

∴ $2ay^2 = 0 \Rightarrow y = 0$ ∴ pt. $(0,0)$

∴ Curve meets x-axis & y-axis at pt. $(0,0)$

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(4) Asymptotes:

→ Coefficient of highest power terms of $x=0$

$-1 \neq 0$

∴ There is no asymptote parallel to x-axis.

→ Coefficient of highest power terms of $y=0$

$2a - x = 0 \Rightarrow x = 2a$

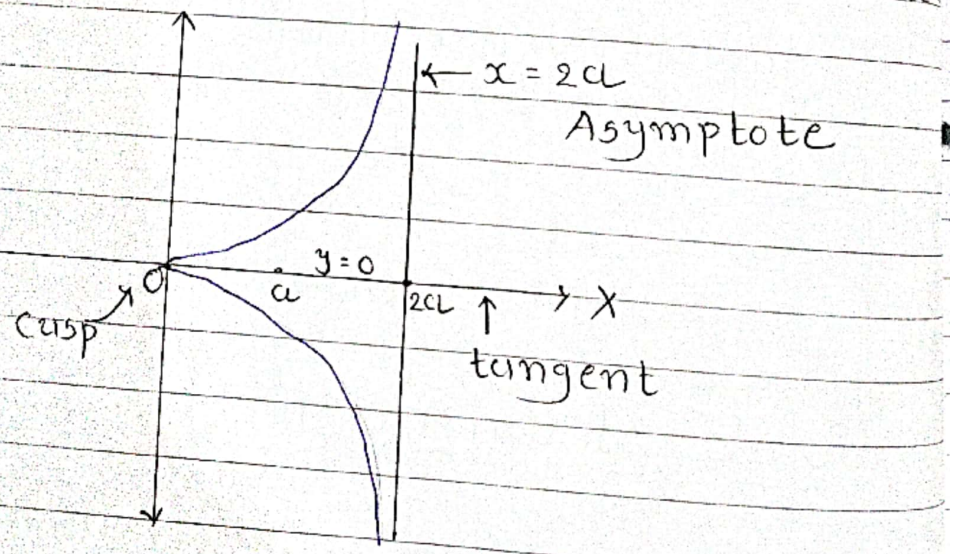
∴ $x = 2a$ is only asymptote parallel to y-ax

(5) Region:

From (i), $y^2 = \frac{x^3}{(2a-x)}$

if we take $x > 2a$ or $x < 0$, then y becomes imaginary.

∴ The curve doesn't exist for $x > 2a$ & x



(Cissoid)

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Ex: 2 Trace the curve $y^2(a+x) = x^2(a-x)$

Sol^{n}:

we have, $y^2 = \frac{x^2(a-x)}{(a+x)}$ —— (i)

$\Rightarrow ay^2 + xy^2 - ax^2 + x^3 = 0$ — (ii)

(1) Symmetry:

The eq^{n} contains only even powers of y.

∴ The curve is symmetric about x-axis.

(2) Origin:

→ eq^{n} (ii) is satisfied by $x=0, y=0$

∴ curve passes through the origin.

→ Tangent at the origin

lowest degree terms = 0

$\Rightarrow ay^2 - ax^2 = 0$

$\Rightarrow y^2 = x^2$

$\Rightarrow y = \pm x$ — tangents are real & distinct

∴ origin is node.

(3) Points of intersection:

→ For pts. on x-axis, put $y=0$ in eq^{n} (ii)

∴ $-ax^2 + x^3 = 0$

$\Rightarrow x^2(x-a) = 0$

$\Rightarrow x = 0, x = a$

∴ Curve intersects x-axis at pts. (0,0) & (a,0)

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→ For pts. on y-axis,

put $x = 0$ in eq^{n} (ii)

$\therefore ay^2 = 0 \Rightarrow y = 0$

$\therefore$ curve intersects y-axis at pt. $(0, 0)$.

(4) Asymptote :

→ coefficient of highest power terms of $x = 0$

$1 \neq 0$

$\therefore$ There is no asymptote parallel to x-axis.

→ coefficient of highest power terms of $y = 0$

$a + x = 0$

$\Rightarrow x = -a$

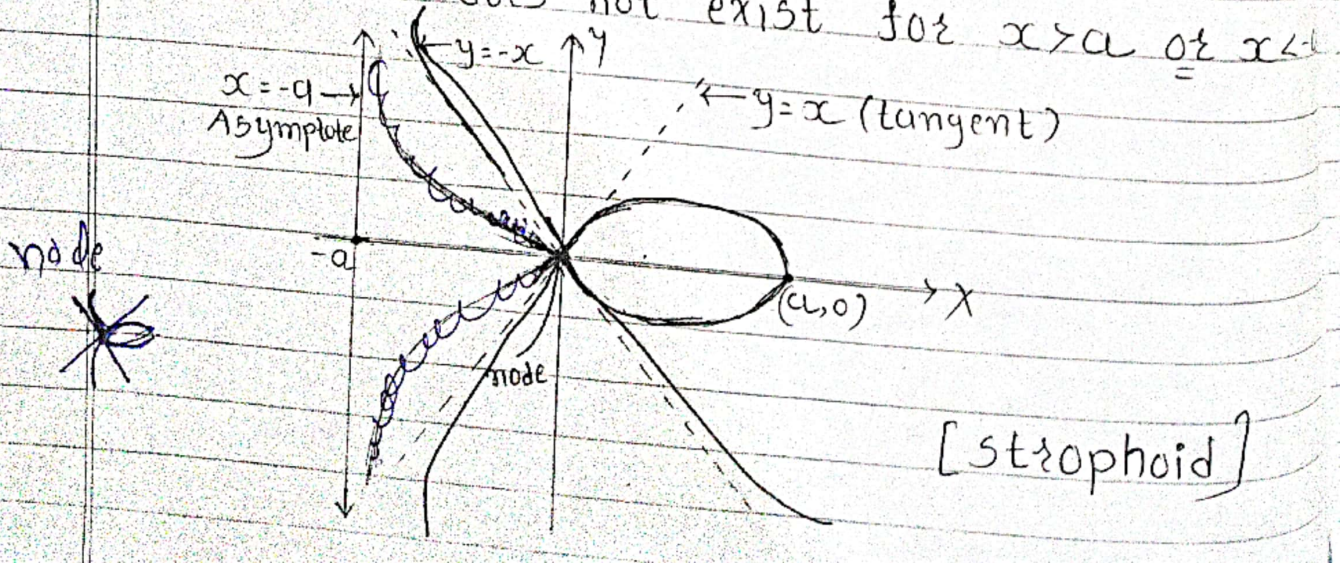
$\therefore x = -a$ is only asymptote parallel to y-axis

(5) Region :

From (i), $y^2 = \dfrac{x^2(a-x)}{(a+x)}$

if we take $x > a$ or $x < -a$ then y become imaginary.

$\therefore$ The curve does not exist for $x > a$ or $x < -a$



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3. Trace the curve $x^3 + y^3 = 3axy$

Sol^n:

We have $x^3 + y^3 = 3axy$ —— (i)

$\Rightarrow x^3 + y^3 - 3axy = 0$ —— (ii)

(1) Symmetry:

If we interchange x and y then eq^n remains same.

∴ The curve is symmetric about the line $y = x$.

(2) Origin:

eq^n (ii) is satisfied by substituting $x = 0, y = 0$

∴ The curve passes through the origin.

→ Tangent at the origin

lowest degree terms $= 0$

$-3axy = 0$

$\Rightarrow x = 0, y = 0$ – tangents real & distinct.

∴ Origin is node.

(3) Points of intersection

→ For pts. on x-axis,

put $y = 0$ in eq^n (ii)

∴ $x^3 = 0 \Rightarrow x = 0$ ∴ pt. $(0,0)$

→ For pts. on y-axis,

put $x = 0$ in eq^n (ii)

$y^3 = 0 \Rightarrow y = 0$ ∴ pt. $(0,0)$

∴ The curve intersects x-axis & y-axis at pt. $(0,0)$ only.

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→ The curve is symmetric about the line $y = x$

∴ put $y = x$ in eq^{n} (ii), we get

$$x^3 + x^3 - 3ax^2 = 0 \Rightarrow 2x^3 - 3ax^2 = 0$$

$$\Rightarrow x^2(2x - 3a) = 0$$

$$\Rightarrow x = 0, \quad x = 3a/2$$

∴ The curve intersects $y = x$ line at pts. $(0,0)$ & $(3a/2, 3a/2)$

(4) Asymptotes :

There is no asymptote parallel to x-axis or y-axis.

→ For oblique asymptote : $y = mx + c$

Highest degree $= n = 3$, Highest degree term $= x^3 + y^3$

Put $x = 1$ and $y = m$

$$\therefore \phi_3(m) = x^3 + y^3 = 1 + m^3$$

$$\phi_3'(m) = 3m^2, \quad \phi_2(m) = -3axy = -3am$$

To find value of m

Take $\phi_3(m) = 0$

$$\Rightarrow 1 + m^3 = 0$$

$$\Rightarrow (m+1)(m^2 - m + 1) = 0$$

$$\Rightarrow m = -1 \quad (\because \text{other roots are imaginary})$$

$$\phi_2(m) = -3am = -3a(-1) = 3a$$

Now,

$$c = -\frac{\phi_2(m)}{\phi_3'(m)} = -\frac{3a}{3m^2} = -\frac{a}{(-1)^2} = -a$$

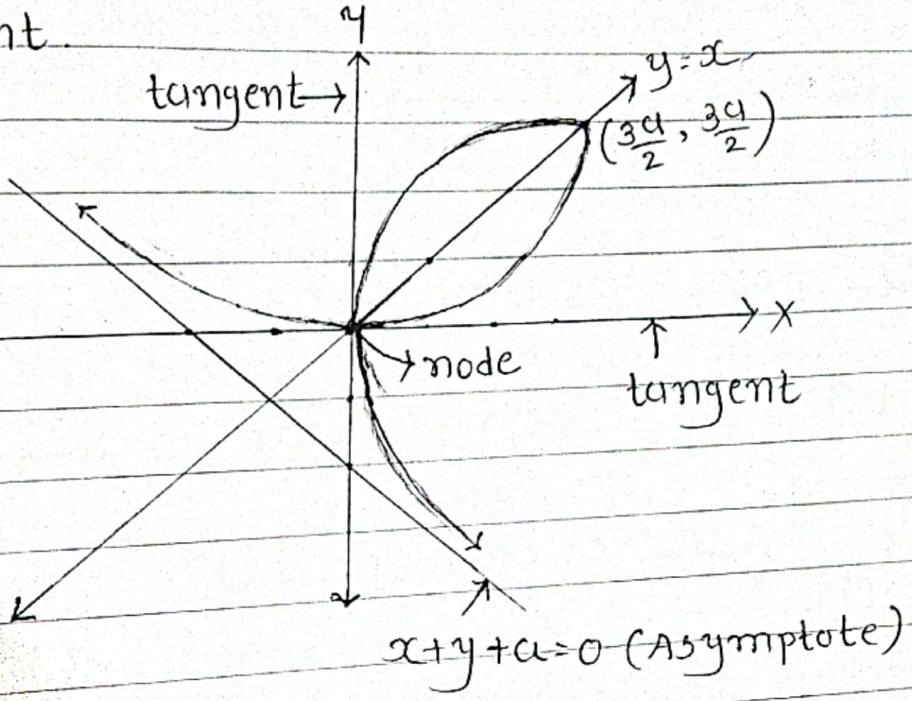
∴ The oblique asymptote : $y = -x - a$

$$\Rightarrow y + x = -a$$

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Region :

Both x & y cannot be negative ~~otherwise~~ because L.H.S. of (i) becomes -ve and the R.H.S. of (i) becomes +ve.
Thus, the eq^{n} of curve is not satisfied.
Hence, the curve doesn't lie in the third quadrant.



[Folium of Descartes]

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